

Day 30 — K-Nearest Neighbors

Veda Technology · Level 2 · Day 30 · Data Science Track

Description: Build a KNN classification model and evaluate how different K values influence prediction performance.

Objective: Understand distance-based classification and the importance of feature scaling.

Dataset: Iris Dataset

1. Workflow

- Load data
- Train-test split
- Feature scaling
- Train KNN
- Test K=1 to 20
- Compare accuracy
- Select best K
- Classification report and confusion matrix

2. Feature Scaling

KNN uses distances, so feature scaling prevents large-range features from dominating.

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

3. KNN Model

```
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)
```

4. K Selection

K values from 1 through 20 are tested. The best K is selected using the highest testing accuracy.

5. Interpretation

Small K: may overfit and react strongly to noise.

Large K: may underfit by smoothing away local patterns.

6. Interview Questions

How does KNN classify? It uses the majority class among the K nearest observations.

Why scale? Because KNN relies on distance calculations.

Too-small K? Possible overfitting.

Too-large K? Possible underfitting.

Parametric? KNN is generally non-parametric.

7. Conclusion

The project demonstrates a complete KNN workflow with scaling, K-value experimentation, best-K selection, and evaluation.